

Md. Ismail Hossain

(575)-551-8706 | statistician71@gmail.com | [linkedin.com/in/md-ismail-hossain](https://www.linkedin.com/in/md-ismail-hossain) | github.com/Ismail-therap

Data Scientist with dual MS degrees in Applied Statistics and Applied Mathematics and 8+ years of experience building machine learning models, NLP pipelines, and AI-driven solutions across healthcare, biomedical research, and industry. Co-authored 8+ peer-reviewed publications (2024–2025), including a lead role on a *Nature Mental Health* study, applying advanced ML and statistical methods to large-scale clinical data. Actively building agentic AI and LLM-based applications (LangChain, OpenAI APIs); completing Stanford School of Medicine’s *AI in Healthcare* (Coursera).

EXPERIENCE

Data Scientist / Analyst <i>University of Wyoming</i> - Applied Prophet time series modeling to 5+ years of indirect cost data for revenue forecasting; used regression modeling to benchmark institutional R&D performance nationally. - Built automated R and Python reporting pipelines; designed interactive Power BI dashboards integrating multi-source institutional datasets.	Dec 2024 – Nov 2025 <i>Laramie, Wyoming</i>
Biostatistician / ML Researcher <i>Louisiana State University Health Shreveport</i> - Applied ML-based biomarker identification and advanced statistical models to clinical datasets; lead analyst on a Nature Mental Health publication and 8+ peer-reviewed studies. - Developed a bioinformatics metabolomics pipeline leveraging ML-based feature selection, reducing analysis costs by \$20,000 per dataset. - Utilized large-scale national datasets (NHANES, NIS, HCUP) for population-level ML and survival analysis across addiction and cardiovascular research.	Sep 2023 – Nov 2024 <i>Shreveport, Louisiana</i>
Graduate Research & Teaching Assistant <i>New Mexico Institute of Mining and Technology</i> - Conducted independent research in statistical modeling and operations research; developed and tested ML models as part of MS Mathematics thesis work. - Assisted in teaching graduate-level statistics and mathematics; supported course design and student mentoring.	Aug 2021 – July 2023 <i>Socorro, New Mexico</i>
Senior Data Scientist <i>Insightin Technology Bangladesh Ltd.</i> - Engineered an automated Python data quality pipeline for ~20GB datasets, achieving a 12× runtime speedup (3 hrs → 15 min) with automated PDF reporting. - Built and deployed NLP sentiment analysis models on large-scale customer review data, contributing to a 5% revenue increase by surfacing actionable dissatisfaction signals. - Led R&D of new data analytics products; explored deep learning architectures for text classification tasks.	Dec 2018 – Jul 2021 <i>Dhaka, Bangladesh</i>
Early Career Data Roles <i>Innovision Consulting & Therap BD</i> Data management in R and SurveyCTO; SQL analysis; BI dashboard QA and real-time Power BI survey monitoring.	2016 – 2018 <i>Dhaka, Bangladesh</i>

FREELANCE CONSULTING

Statistical Consultant (Upwork / PeoplePerHour) <i>ML modeling, statistical analysis, and data science solutions for 193 international clients</i>	July 2013 – July 2021 <i>Remote</i>
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EDUCATION

New Mexico Institute of Mining and Technology <i>MS Mathematics (Operations Research & Statistics)</i>	Socorro, NM <i>July 2023</i>
University of Dhaka <i>MS Applied Statistics BS Applied Statistics</i>	Dhaka, Bangladesh <i>Dec 2015 Dec 2013</i>

SELECTED PROJECTS

Intrusion Detection in Industry 4.0 using ML (2023) Trained Random Forest, SVM, and unsupervised models (K-means, SOM) on IoT telemetry data; Random Forest achieved 99.4% accuracy – published as a preprint (Authorea, 2023).
NLP: Sentiment Prediction on Customer Reviews (2022) - Compared LSTM, CNN, and CNN+LSTM in Python for multi-class sentiment classification; CNN achieved 91.5% (binary) and CNN+LSTM 73.4% (3-class). - Designed an interactive Power BI dashboard to visualize sentiment trends and model outputs, enabling non-technical stakeholders to act on customer insights directly.

SKILLS

ML & AI: Scikit-learn, TensorFlow, PyTorch, LSTM, CNN, Random Forest, XGBoost, LangChain, OpenAI API, HuggingFace, AzureML.
Data & Engineering: Python, R, SQL, PySpark, AWS, Azure Blob Storage, Google Colab, Git.
Methods: NLP, Deep Learning, Survival Analysis, Bayesian Methods, TMLE, Propensity Score Matching, Time Series, Causal Inference.